Supplementary Material for

AI Should Formulate, Not Negotiate: LLM-Formulated Target Cascading for Space-Claim Arbitration in Compact Product Design

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This document gives the evaluator equations and constants, the acoustic model, the settings of every method, the full prompts with the reference product's name removed, per-repeat results, example decision records, data sources, software versions, and a log of protocol changes made during the study. Equation, table, and section numbers prefixed with "S" refer to this document; others refer to the main paper.

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S1 Benchmark Constants

The interior volume constraint is

$$V_b + 2.24A_{pcb} + V_s + V_a + V_{ac} + V_h + 0.78A_t + V_l + V_f \leq 11{,}200 - 1500\,(t - 1.2). \tag{S1}$$

Board and spreader volumes are their areas times the fixed thicknesses above. Cell thickness is V_b/A_b.

S2 Domain Evaluators

All evaluators are closed form. Units are mm, mm^2, mm^3, W, and °C unless stated.

Table S1: Shared pools and fixed constants

Symbol	Meaning	Value
V_{pool}	Contested interior volume at $t = 1.2$ mm	11,200 mm^3
$\partial V/\partial t$	Interior volume lost per mm of added wall	1500 mm^3/mm
V_{R6}	Perimeter non-conductive keepout (antenna + NFC loop)	2400 mm^3
A_{R2}	Caseback plane outside the charging-coil exclusion	1143 mm^2
A_{R7}	Total through-wall opening (speaker port + 1 mm^2 microphone)	12 mm^2
z_{avail}	Height between sensor stack and display	7.1 mm
t_{pcb}	Board plus component stack (10 layers, fixed)	2.24 mm
t_{spr}	Effective spreader thickness (graphite + TIM)	0.78 mm
V_{drv}	Driver envelope (18×13×2.5 mm class)	540 mm^3
ρ_v	Cell volumetric energy density	600 Wh/L
P_{avg}	Typical-use average load	38 mW
P_{sys}	Background load during siren	0.15 W
δ, η_{amp}	Siren duty cycle, amplifier efficiency	0.5, 0.7
P_{soc}	Sustained processor power for the skin case	0.9 W

Battery. $E = \rho_v V_b$ (Wh, with V_b in liters); runtime $= E/P_{\text{avg}}$; siren endurance from Eq. (1) of the main paper.

Antenna (GNSS L5). With clearance volume V_a treated as a sphere of radius $a = (3V_a/4\pi)^{1/3}$ and $k = 2\pi f/c$ at $f = 1176.45$ MHz, the Chu bound [4] gives

$$Q_{\min} = \frac{1}{(ka)^3} + \frac{1}{ka}, \qquad \eta_{L5} = \min\left(1, \frac{f/Q_{\min}}{B}\right) \min\left(1, \frac{d_c}{1.5}\right) \left(\frac{0.8}{t}\right)^{1/2}, \tag{S2}$$

with required bandwidth $B = 24$ MHz and conductor distance $d_c = V_a/(167 \times 6)$ mm (perimeter 167 mm, band height 6 mm). The last factor detunes the antenna as the titanium wall thickens.

NFC (13.56 MHz). Loop area $A_\ell = 400\,V_l/320$ mm^2, ferrite thickness $t_f = V_f/900$ mm, distance to titanium $d = 1$ mm, four turns, $\mu_r = 120$:

$$r_{\text{NFC}} = 22\sqrt{A_\ell/400}\left[1 - 0.85\,\frac{e^{-d/2}}{1 + \mu_r t_f/12}\right] \text{ mm} \geq 10. \tag{S3}$$

Haptics. Linear resonant actuator at 170 Hz with 0.35 mm stroke, moving-mass density 17 g/cm^3 and mass fraction 0.45: $F = 0.45\,\rho V_h\,(2\pi \cdot 170)^2\,x \geq 0.9$ N.

Thermal. $T_{\text{skin}} = 25 + P_{\text{soc}}\,(5 + 1280/A_t + R_{\text{case}} + 0.15\,t) \leq 43$, with $R_{\text{case}} = 10.82$ K/W calibrated so that the nominal spreader ($A_t = 320$ mm^2, $t = 1.2$ mm) sits at 43.0 °C.

Structure (10 ATM). Ports act as stress raisers and ribs add stiffness:

$$m = 1.55\,(t/1.2)^{1.5} + 0.15\,V_s/1800 - 0.0625\,a_p + 0.30 \geq 1.5. \tag{S4}$$

The coefficients were set so that an 8 mm^2 port at nominal wall and ribs gives exactly the required margin.

Board. Routability $= \text{clip}\big((A_{pcb}/1400)/0.9, 0, 1.5\big) \geq 0.95$ for 1400 pins at 10 layers. This is the weakest evaluator and is flagged as a limitation in the main paper.

Mass (reported, not constrained). $m = 15.9 + 3.04 \times 10^{-3} V_b + 24.0 + 6.6(t - 1.2) + 2.7 \times 10^{-3}(V_s - 1800) + 5.14 \times 10^{-3} V_h + 1.12 \times 10^{-3} V_{ac} + 5.0 \times 10^{-3} V_f + 4.0 A_{pcb}/1250$ g.

S3 Acoustic Model

S3.1 Circuit

The driver is coupled through a gyrator to an acoustic impedance network (rms quantities, angular frequency ω). With electrical impedance $Z_e = R_e + j\omega L_e$, driver acoustic impedance $Z_{ad} = (R_{ms} + j\omega M_{ms} + 1/j\omega C_{ms})/S_d^2$, back cavity

$Z_b = \rho c^2/(j\omega V_{\text{back}})$, and source pressure $p_s = V\,Bl/(Z_e S_d)$, the driver volume velocity is

$$U_d = \frac{p_s}{Bl^2/(Z_e S_d^2) + Z_{ad} + Z_b + Z_{\text{front}}}, \qquad Z_{\text{front}} = \left(j\omega C_f + \frac{\omega C_f}{q_{th}} + \frac{1}{Z_{br}} \right)^{-1}, \tag{S5}$$

where $C_f = V_{\text{front}}/\rho c^2$, q_{th} represents thermal loss in a front cavity of 1 mm effective height, and the outlet branch is

$$Z_{br} = R_{\text{duct}} + R_{\text{rad}} + \frac{r_m}{a_p} + R_{nl} + j\omega \frac{\rho\, \ell_{\text{eff}}}{a_p}. \tag{S6}$$

The outlet is a $w \times h$ rectangle of area a_p and length 3 mm; h above $\sqrt{a_p}$ is treated as square. Viscous loss uses the boundary-layer thickness $\delta = \sqrt{2\nu/\omega}$: $R_{\text{duct}} = \rho\omega\delta\,\ell_r P/(2a_p^2)$, with perimeter P, $\ell_r = \ell + 0.5 d_h$, and $\ell_{\text{eff}} = \ell + 0.85 d_h$ for hydraulic diameter d_h, consistent with the wide-duct limit of [5]. Radiation resistance is $R_{\text{rad}} = \rho\omega^2/2\pi c$, and the nonlinear orifice term is $R_{nl} = C_{nl}\rho|U_{\text{out}}|/a_p^2$ with $C_{nl} = 1$, iterated at the operating level. The outlet flow is $U_{\text{out}} = U_d Z_{\text{front}}/Z_{br}$, and the half-space far-field pressure at distance r is $|p| = \rho\omega|U_{\text{out}}|/2\pi r$. Electrical power is $\text{Re}(VI^*)$. The free acoustic volume $V_{ac} - 540$ mm^3 is split into back (s) and front ($1-s$) cavities. The main case is an open, water-tolerant cavity with a sealed driver, so no membrane is in the path.

S3.2 Driver Parameters

Table S2: Driver parameters (18×13 mm class)

Parameter	Value	Basis
R_e, L_e	7.2 Ω, 70 μH	datasheet
f_s (free air), Q_{ms}	500 Hz, 2.4	datasheet
S_d	157 mm^2	datasheet
C_{ms}	0.753 mm/N	from the datasheet resonance shift in a 1 cm^3 box
M_{ms}	134.6 mg	from f_s and C_{ms}
R_{ms}	0.176 N·s/m	from Q_{ms}
Bl	1.086 T·m	fitted to 95 dB at 1 W, 10 cm (datasheet typical)
Rated power	1.0 W	datasheet

The datasheet's electrical Q (5.2–6.3) corresponds to its 92 dB minimum sensitivity rather than the 95 dB typical value. At 92 dB no design in the benchmark is feasible; the main case uses the typical value.

S3.3 Siren and Voice Metrics

Siren. Five tones between 3.0 and 3.4 kHz are evaluated at 1 V. For the tone with the highest sensitivity, the drive is scaled to give 86 dB at 1 m, R_{nl} is updated from the resulting outlet flow, and the process repeats until the flow changes by less than 0.2%. The constraint is the electrical tone power $P_{\text{siren}} \le 1.0$ W. Resonance Q is the peak frequency divided by the -3 dB bandwidth of the small-signal response between 1.5 and 14 kHz, and is limited to 15.

Voice. 100 × SII by the ANSI S3.5-1997 third-octave procedure [1] (MOSQITO 1.2.1 implementation). Far-end speech with the ANSI normal-effort spectrum shape is replayed at 0.1 W rms total electrical power, distributed across bands by that spectrum. The listener is at 0.3 m, the noise is speech-shaped at 55 dB SPL overall, and hearing is normal. Band speech levels follow from the model's sensitivity at each band center. An earlier proxy based on low-frequency corner and ripple saturated whenever the corner exceeded 1 kHz and was replaced before any of the reported runs.

S4 Method Settings

A1 waterfall. Order: board, battery, antenna, thermal, structure, haptics, NFC, acoustics. Board takes 1250 mm^2; battery takes the volume implied by the program's runtime target with an 800 mm^2 footprint, or the thickest cell

allowed if needed; antenna takes the program allocation; thermal takes the smallest spreader meeting 43 °C at the current wall; structure takes 1800 mm^3 of ribs and the thinnest wall that meets the pressure margin at the interface-specification port of 9 mm^2; haptics takes 350 mm^3; NFC takes 320/180 mm^3. Acoustics may use up to the remaining volume (900–2400 mm^3), meets the siren power limit first, then maximizes voice. Rework rules and the 50-program sweep are as in the main paper. Mean rework loops were 5.9 (maximum 12); at runtime targets of 55–62 h the mean was 9.3.

A2 NSGA-III. pymoo 0.6.2, population 120, 120 Das–Dennis reference directions (7 partitions), 1500 generations, seeds 0–4, default operators. Constraints handled by pymoo's feasibility-first rule.

A3 hand ATC. Fourteen arbiter targets (all allocations except the acoustic internals, plus siren tone power and voice); sixteen local copies across eight subproblems; arbiter objective a weighted sum with normalization ranges runtime 40–63 h, siren 1.5–3.5 h, voice 70–85, L5 0.05–0.30; $\beta = 2.2$, $\gamma = 0.4$, $w_0 = 1$, $v_0 = 0$, tolerance 10^{-3}, 40 outer iterations; SLSQP for the arbiter and seven subproblems, differential evolution then SLSQP for acoustics. Mean outer iterations 9.4; one run reached the limit. Wall-clock 129 s for 35 runs.

A4 negotiation. Speaking order: board, battery, antenna, thermal, structure, acoustics, haptics, NFC. Shared inputs each agent may probe read-only: structure–port area; thermal and antenna–wall thickness; acoustics–battery volume. Probe values outside bounds are clipped and the clipping is reported to the agent. Maximum 12 rounds; 120-word messages; the moderator reports the ledger (pool totals and over-commitment) and notices. Output limit 1200 tokens per turn.

A1-PM escalation. After each waterfall pass whose design meets every hard constraint but gives less than 3 h of siren endurance, the PM holds a review using only what the leads report: the siren tone power P and the battery energy needed for 3 h, $E_{\mathrm{req}} = 3(\delta P/\eta_{\mathrm{amp}} + P_{\mathrm{sys}})$. Levers, one per review: E1 acoustics minimizes P subject to voice ≥ 70 and $Q \leq 15$; E2 battery specification raised to $1.02\,E_{\mathrm{req}}/\rho_v$ (repeated while the battery still grows by more than 0.5%); E3 interface port raised to 11 mm^2; E4 acoustic volume reserved at 1300 mm^3, then $+100$ mm^3 per review. At most six reviews. A rework loop counts each waterfall iteration across all passes; each review adds one.

A4-PM. As A4, plus a PM agent that speaks last each round, sees the integrated objectives, failing constraints, and siren tone power of the current claims, may request three what-if studies of complete allocations per round, and may issue binding directives on any variable. Agreement requires eight ACCEPTs, balanced pools, and PM APPROVE; a PM APPROVE is recorded as CONTINUE if the integrated design fails any hard constraint. Output limit 4000 tokens per turn for all agents (A4 used 1200; about 3% of A4 turns appear to have been cut short by that limit).

A5 engine. As A3, with the arbiter objective and floors from the LLM's formulation; the siren floor split across battery and acoustics (Eq. (4) of the main paper); runtime floors, when present, enforced at the arbiter; achievement function with $\kappa = 40$, $\rho = 0.05$; restoration by SLSQP with the full model; 2% movement limit. The coordination parameters are fixed and not offered to the LLM.

S5 Prompts

The prompts are reproduced verbatim except that the reference product's name and processor part number are replaced by "[withheld]".

S5.1 Shared Program Brief (A4 and A5)

```
You are one engineering lead in the product-architecture review for a rugged 47 mm titanium smartwatch (
    reference platform: [withheld]). The case envelope is frozen by ID. All internal components must share
    11,200 mm^3 of interior volume at a 1.2 mm wall (each +0.1 mm of wall costs everyone 150 mm^3). Other
    shared pools: caseback plane 1143 mm^2 (battery footprint + thermal spreader), perimeter keepout 2400 mm
    ^3 (antenna + NFC loop; the keepout lies INSIDE the 11,200 mm^3 interior, so antenna and NFC volumes are
    counted once, in the interior total, and the keepout is an extra limit on those two only - nothing is
    double-counted), through-wall openings 11 mm^2 for the speaker port. Program goals, all important: ~60 h
    battery runtime, an emergency siren of 86 dB at 1 m that keeps sounding for >= 3 hours (endurance depends
     on battery energy and siren power), good voice intelligibility, and strong GNSS L5 reception. The
    current claims over-commit the interior. You must reach agreement with the other leads. Be concise and
    concrete; argue from your numbers.
```

S5.2 Domain Requirements (A4 agents and A5 formulation)

```
- board: 10-layer board, [SoC withheld]. Routability index must stay >= 0.95 (about 1197 mm^2 minimum). Board
    volume = area x 2.24 mm.  [variables: A_board]
- battery: Runtime target ~60 h typical use (marketing). Cell thickness 2.6-4.8 mm; battery + 2.24 mm board
    stack <= 7.1 mm. Battery footprint and the thermal spreader share the 1143 mm^2 caseback plane.  [
    variables: V_batt, A_batt]
- antenna: GNSS L1+L5, LTE, Wi-Fi, BT. Non-conductive perimeter keepout 900-2400 mm^3; shares the 2400 mm^3
    perimeter with the NFC loop. Thicker titanium walls detune you. Maximise L5 efficiency.  [variables:
    V_ant]
- thermal: Skin temperature <= 43 C at 0.9 W sustained. Spreader area 180-520 mm^2 on the caseback (volume =
    area x 0.78 mm). Depends on wall thickness.  [variables: A_spr]
- structure: Titanium case, 10 ATM and MIL-STD-810H. Pressure margin >= 1.5 including the speaker port opening
    . Ribs 1400-2400 mm^3, wall 1.0-1.8 mm. Each extra 0.1 mm of wall removes ~150 mm^3 of interior volume
    from everyone.  [variables: V_struct, t_wall]
- acoustics: Single 18x13 mm driver (540 mm^3 envelope), open water-tolerant cavity, hydrophobic mesh >= 20
    rayl. Emergency siren 86 dB @ 1 m with <= 1.0 W driver power, gapped 50% duty; voice intelligibility (SII
    ). Volume 900-2400 mm^3; port 3-11 mm^2 (through the structure's wall); split = back-cavity share of free
     volume.  [variables: V_acou, port, split, h_out, r_mesh]
- haptics: LRA force >= 0.9 N (about 295 mm^3 minimum).  [variables: V_hap]
- nfc: NFC read range >= 10 mm. Loop 180-520 mm^3 in the perimeter keepout, ferrite 110-300 mm^3 inside.  [
    variables: V_loop, V_ferr]
```

S5.3 A4 Protocol Rules (final protocol)

```
PROTOCOL: Each round every lead speaks once, in a fixed order, seeing what was said earlier that round. Reply
    with at most 120 words to the other leads, then one fenced JSON block: {"claims": {<your variables>}, "
    probe": [<up to 3 alternative dicts>], "status": "ACCEPT" or "OBJECT"}. Probe dicts may contain your own
    variables and, as read-only what-ifs, the shared inputs listed for you; out-of-bound values are clipped
    and reported. ACCEPT means BOTH (a) your own requirements are met in your numbers and (b) you can live
    with the whole ledger. If your own requirement fails, you must OBJECT. When everyone accepts and the
    pools balance, claims are frozen for a CONFIRMATION round; any REJECT reopens the negotiation.
```

S5.4 A4-PM Program-Manager Prompt

The PM receives the shared brief and rules, followed by:

```
YOU ARE THE PROGRAM MANAGER. You own the product requirements and have decision authority over every
    allocation. You see the integrated system numbers of the current claims, which no lead sees, and you may
    request up to 3 what-if studies of complete allocations per round (results arrive next round). You may
    issue binding DIRECTIVES that fix any variable; a lead cannot change a directed variable until you change
     or release it (set it to null to release). Goal: an agreed design that meets EVERY hard requirement,
    including siren >= 3 h, with the program direction as priority. Reply with at most 150 words to the leads
    , then one fenced JSON block: {"directives": {<variable>: value or null}, "probe": [<up to 3 dicts of
    variable values>], "status": "APPROVE" or "CONTINUE"}. APPROVE only if the current design meets every
    requirement.
```

and then the ledger, all current claims, variable bounds and owners, the integrated system numbers, its active directives, the results of its what-if studies from the previous round, and the leads' messages from the current round.

S5.5 A5 Formulation Prompt

System prompt:

```
You are a senior systems engineer setting up the arbiter of a multidisciplinary design optimisation.
```

User prompt, in order: the requirements (as above), the probe results as JSON (Table S4), the program direction if any, the baseline formulation if revising, then the catalog and instructions:

```
ARBITER FORMULATION CATALOG (Analytical Target Cascading).
Fixed by the domains (you do not change these): the arbiter sets 14 targets - every domain's allocation
(battery volume/footprint, board area, structure ribs and wall thickness, antenna keepout, acoustic volume,
port area, haptics, thermal spreader, NFC loop and ferrite) plus two acoustic RESPONSES (siren tone power,
voice SII). Each domain holds its own constraints and matches the targets as closely as it can; an
augmented-Lagrangian loop makes arbiter and domains consistent. Interior volume, caseback, perimeter and
port pools are always enforced at the arbiter.

System objectives the arbiter can evaluate (all maximised):
  runtime : battery runtime, hours (physically ~41 to ~62 h in this envelope)
  siren   : emergency-beacon endurance in hours at 86 dB @ 1 m, 50% duty (depends on battery energy AND siren
     power)
  voice   : voice intelligibility, 100 x ANSI SII (roughly 70 to 90 for feasible designs)
  l5      : GNSS L5 antenna efficiency, 0..1 (roughly 0.05 to 0.30 for feasible designs)

YOU DECIDE (return as JSON):
  "normalisation": {obj: [worst_acceptable, best_plausible]} for all four objectives. Scales each objective to
                   0..1 for the trade; choose from the requirements, not to flatter any one objective.
  "scalarisation": "weighted_sum" or "achievement". weighted_sum maximises sum(w*n); it cannot reach
                   non-convex parts of the trade-off. achievement minimises the largest weighted shortfall
                   from an aspiration point (reference-point / Tchebycheff method) and can reach any Pareto
     point.
  "aspiration":    {obj: value} required if scalarisation is achievement - what the program is aiming for.
  "objective_floors": {obj: minimum} - program statements that are REQUIREMENTS rather than goals become hard
                   constraints at the arbiter. Use {} if none. Floors that cannot be met make every run
     infeasible.
  "weights":       {obj: w} default priorities (they will be normalised).
  "justification": {field: one sentence citing the requirement or probe result that drove it}
Solver parameters (penalty weights, tolerances, iteration limits) are owned by the solver and are not yours to
      set.

Set aspirations and normalisation from the probe results so that every aspiration is reachable or only
     slightly beyond what was achieved; an unreachable aspiration makes the weights ineffective.
Return ONLY one fenced ‘‘‘json block with keys normalisation, scalarisation, aspiration, objective_floors,
     weights, justification.
```

If validation fails, the model is shown the errors and its previous answer once and asked again.

S5.6 A5 Decision-Record Prompt

System prompt:

```
You are the program's systems engineer writing the decision record for an architecture review. Explain the
     allocation to engineers who will challenge it. Use ONLY the numbers provided; do not invent values. Cite
     numbers exactly as given.
```

User prompt:

```
DATA:
<JSON: program_direction, formulation (scalarisation, objective_floors, weights), objectives_achieved,
     allocation, each_domains_original_claim, binding_or_near_binding_constraints, acoustic_details,
     coupling_prices, price_meaning>

Write the decision trace in at most 250 words: (1) which domains conceded what versus their original claims,
     (2) which couplings drove those concessions, citing the coupling prices, (3) what the design gives up,
     (4) one risk a reviewer should probe.
```

S5.7 Program Directions

S6 Additional Results

The comparison uses the design each direction produced, whether or not it passed the 2% repair rule.

Table S3: The ten program directions (A4 and A5)

Key	Text
balanced	(none; the stated goals only)
battery_first	Battery runtime is the headline spec; protect ˜60 h first.
siren_first	The emergency siren is the safety-critical feature; it must meet 86 dB @ 1 m and last ¿= 3 h.
gnss_first	Outdoor athletes are the core customer; GNSS L5 reception is the top priority.
voice_first	Call quality reviews were poor last generation; voice intelligibility is the top priority.
mass_light	Keep every domain lean; give back any volume you do not need.
schedule	Schedule is tight; converge quickly and avoid reopening the port or wall.
battery_siren	Battery runtime and siren endurance are joint headline specs.
rf_voice	Connectivity and calls matter most: GNSS and voice first.
no_direction	There is no program priority beyond the stated goals; settle it among yourselves.

Table S4: Probe solves (neutral formulation, no floors; all feasible)

Weights	Runtime (h)	Siren (h)	Voice	L5
equal	55.30	2.642	75.17	0.174
runtime	62.28	2.889	74.63	0.092
siren	62.03	3.141	76.04	0.073
voice	51.42	2.261	77.73	0.153
l5	45.20	2.098	74.61	0.267

Table S8: A4 negotiations, final protocol (objectives: runtime h, siren h, voice, L5)

Rep.	Direction	Rounds	Confirmed	Feasible	RT	Siren	Voice	L5	Violated
1	balanced	12	no	no	59.8	2.57	74.4	0.170	siren driver power 1W
1	battery_first	12	no	no	59.8	3.02	76.1	0.125	C1 volume
1	siren_first	12	no	no	60.0	0.85	83.5	0.133	siren driver power 1W
1	gnss_first	12	no	no	60.0	1.40	81.4	0.217	C1 volume, O6 pressure, siren driver power 1W
1	voice_first	12	no	no	56.8	2.02	83.9	0.119	siren driver power 1W
1	mass_light	12	no	no	60.0	1.25	81.0	0.189	O6 pressure, siren driver power 1W
1	schedule	12	no	no	57.9	1.60	82.4	0.140	C1 volume, siren driver power 1W
1	battery_siren	12	no	no	59.8	0.55	87.4	0.181	siren driver power 1W
1	rf_voice	12	no	no	53.7	1.26	84.8	0.206	siren driver power 1W
1	no_direction	12	no	no	59.6	0.81	83.7	0.185	siren driver power 1W
2	balanced	12	no	no	58.5	1.84	82.8	0.133	siren driver power 1W
2	battery_first	12	no	no	60.2	2.58	73.7	0.145	siren driver power 1W
2	siren_first	12	no	no	59.8	1.79	82.3	0.133	siren driver power 1W
2	gnss_first	12	no	no	55.2	1.66	82.3	0.208	C1 volume, siren driver power 1W
2	voice_first	12	no	no	58.7	1.31	85.3	0.121	C1 volume, siren driver power 1W
2	mass_light	12	no	no	59.7	2.01	78.8	0.157	siren driver power 1W
2	schedule	12	no	no	58.4	1.82	82.4	0.152	siren driver power 1W
2	battery_siren	12	no	no	60.0	1.29	80.9	0.179	siren driver power 1W
2	rf_voice	12	no	no	56.4	2.02	77.1	0.208	siren driver power 1W
2	no_direction	12	no	no	59.3	1.39	84.8	0.167	C1 volume, siren driver power 1W
3	balanced	12	no	no	59.1	1.05	83.5	0.202	siren driver power 1W
3	battery_first	12	no	no	59.8	1.84	79.4	0.150	siren driver power 1W

Rep.	Direction	Rounds	Confirmed	Feasible	RT	Siren	Voice	L5	Violated
3	siren_first	12	no	no	59.8	2.73	74.6	0.145	O6 pressure
3	gnss_first	12	no	no	58.5	1.04	80.4	0.222	siren driver power 1W
3	voice_first	12	no	no	55.5	1.54	85.6	0.162	siren driver power 1W
3	mass_light	12	no	no	60.0	1.07	83.2	0.159	O6 pressure, siren driver power 1W
3	schedule	12	no	no	59.2	2.25	71.1	0.152	C1 volume, siren driver power 1W
3	battery_siren	12	no	no	60.8	1.18	73.2	0.163	siren driver power 1W
3	rf_voice	12	no	no	53.7	1.50	80.3	0.208	siren driver power 1W
3	no_direction	12	no	no	59.2	0.62	80.2	0.161	siren driver power 1W

S6.1 A1-PM Escalation Results

Lever use across all programs: E1 50, E2 80, E3 30, E4 30.

S6.2 A4-PM Negotiation Results

Table S10: A4-PM negotiations (12 rounds each; none confirmed). *P*: siren tone power (limit 1 W).

Direction	Feasible	RT	Siren	Voice	L5	*P* (W)	Final directives	Objectors	$
balanced	no	60.0	2.64	73.7	0.164	1.001	V_ant, V_batt, port, t_wall	1	1.60
battery_first	no	59.8	1.33	84.2	0.133	2.173	A_batt, V_ant, V_batt, V_hap	1	1.42
battery_siren	no	61.3	2.69	73.7	0.156	1.002	V_acou, V_struct, port, t_wall	1	1.66
gnss_first	no	61.6	1.93	69.4	0.211	1.493	A_batt, V_acou, V_batt, r_mesh, split	1	1.58
mass_light	yes	61.0	2.69	74.1	0.156	0.999	V_acou, V_batt, port, t_wall	1	1.54
no_direction	no	60.0	1.63	80.0	0.118	1.752	V_acou, V_ant	8	1.67
rf_voice	no	58.2	2.55	73.7	0.208	1.002	V_batt, t_wall	1	1.56
schedule	yes	61.6	2.73	74.2	0.133	0.991	V_ant, port, r_mesh	8	1.67
siren_first	no	58.3	2.56	79.2	0.080	1.001	A_board, A_spr, V_ant, V_batt, port, t_wall	8	1.72
voice_first	no	59.2	1.45	84.8	0.141	1.969	t_wall	7	1.64

S7 Decision Records

S7.1 Grounding Check

Every numeral in a record is extracted with a regular expression. A number counts as grounded if it matches, within 1% relative or 0.006 absolute, any number in the data the model received, its absolute value, or 100 times it (a percentage); or if it matches a difference, ratio, or percentage formed from a claimant's original claim and its allocation. Zero is skipped. The check is lenient in two ways: list labels such as "1." are counted and usually match, and numbers written in scientific notation are split. A stricter variant that drops section labels and bare integers up to 10 gives 98.1% for Sonnet, 97.7% for Haiku, and 96.5% for Opus, against 98.3%, 97.7%, and 96.8% for the check as defined.

Table S5: Baseline formulations written by each model (aspirations: runtime, siren, voice, L5; normalization ranges in the same order)

Model	Rep.	Scalarization	Floors	Aspiration	Normalization
Sonnet 5.5	1	achievement	{"siren": 3.0}	58.0, 3.05, 76.5, 0.15	45–62; 2–3.2; 72–79; 0.05–0.28
Sonnet 5.5	2	achievement	{"siren": 3.0}	60.0, 3.1, 76.5, 0.11	45–62.5; 2–3.3; 70–80; 0.05–0.28
Sonnet 5.5	3	achievement	{"siren": 3.0}	60.0, 3.1, 77.0, 0.14	45–62.5; 2–3.2; 70–78; 0.05–0.28
Haiku 5.5	1	achievement	{"siren": 3.0}	60.0, 3.14, 77.7, 0.25	41–62; 3–3.5; 70–90; 0.05–0.3
Haiku 5.5	2	achievement	{"siren": 3.0}	60.0, 3.15, 77.8, 0.25	41–62.5; 2–3.2; 70–90; 0.05–0.3
Haiku 5.5	3	achievement	{"siren": 3.0}	62.0, 3.1, 78.0, 0.27	41–62.5; 2–3.2; 70–90; 0.05–0.3
Opus 5.5	1	achievement	{"siren": 3.0}	60.0, 3.1, 76.5, 0.2	50–62.3; 3–3.15; 74.6–77.8; 0.07–0.27
Opus 5.5	2	achievement	{"siren": 3.0}	60.0, 3.05, 76.5, 0.18	45–62.3; 2.1–3.15; 74.6–77.8; 0.07–0.27
Opus 5.5	3	achievement	{"siren": 3.0}	60.0, 3.1, 77.0, 0.2	50–62.3; 3–3.15; 74.6–77.8; 0.07–0.27

Table S6: A5 per repeat (45 runs each: 35-weight sweep and 10 directions)

Model	Rep.	ATC-strict	Feasible ($\leq$2%)	Median repair (%)	Mean outer	At limit	Grounded numbers
Sonnet 5.5	1	7	32	0.66	29.4	27	500/514
Sonnet 5.5	2	17	38	0.08	19.8	13	529/533
Sonnet 5.5	3	11	36	0.34	24.5	19	528/537
Haiku 5.5	1	8	22	2.03	27.3	26	523/526
Haiku 5.5	2	7	25	1.33	31.0	31	476/499
Haiku 5.5	3	6	23	1.99	31.6	31	513/522
Opus 5.5	1	14	37	0.40	25.0	21	474/495
Opus 5.5	2	5	26	1.53	30.9	28	512/540
Opus 5.5	3	16	39	0.02	19.9	14	503/503

S7.2 Example Records

The records below are for the `siren_first` direction, first repeat, one per model. They are reproduced verbatim apart from character substitutions for typesetting (arrows written as `->`). Each model saw its own formulation and solved design, so the numbers differ between them.

Sonnet 5.5 (301 words, 51/51 grounded):

```
**Decision record: allocation under the siren-first direction**

**1. Concessions versus original claims**
- Antenna took the largest cut: V_ant 1500 -> 966.697.
- V_struct fell 1800 -> 1489.885.
- V_acou fell 1460 -> 1267.164.
- V_hap fell 350 -> 299.659, V_loop 320 -> 269.833, and V_ferr 180 -> 164.334.
- V_batt fell 3786 -> 3657.9, and A_board fell 1250 -> 1197.
- A_batt held at 800.
- Some variables moved up or sideways: t_wall rose 1.2 -> 1.302, port rose 9.0 -> 10.808, A_spr rose 320 ->
    321.226, split dropped 0.78 -> 0.3, h_out rose 1.0 -> 3.5, and r_mesh dropped 40 -> 20.

**2. Couplings that drove it**
- bat:V_batt at -21728.57 dominates everything else. Battery volume was the hardest coupling to settle, and it
     is tied to the 3 h siren floor.
- Next are str:t_wall (-635.31), board:A_board (-449.61), ac:V_acou (-324.76), and ac:P_siren (-268.06).
- str:port (+196.77) pushed the opposite way, which is consistent with port moving up to 10.808.
- The antenna, haptic, and NFC prices are all 0.0. Those domains gave up volume with no price signal, so the
    arbiter never tested how much the cuts hurt them.

**3. What the design gives up**
- The siren sits exactly at its floor of 3.0, so it has no slack.
- The other objectives are runtime 57.756, voice 76.357, and l5 0.083.
- Binding or near-binding limits are C1_volume, O6_pressure, O5_skin_temp, haptic_force, and board_routability
    .
```

Table S7: Objective-first directions that improved their named objective over the balanced direction

Model	Rep.	Count	Directions honored
Sonnet 5.5	1	3	battery, voice, gnss
Sonnet 5.5	2	3	battery, voice, gnss
Sonnet 5.5	3	4	battery, siren, voice, gnss
Haiku 5.5	1	2	battery, voice
Haiku 5.5	2	2	voice, gnss
Haiku 5.5	3	3	battery, voice, gnss
Opus 5.5	1	2	battery, voice
Opus 5.5	2	3	battery, voice, gnss
Opus 5.5	3	2	battery, siren

Table S9: A1-PM by runtime target (five antenna allocations each): reviews and rework loops per program

Runtime target (h)	Reached 3 h	Reviews	Rework loops	Siren (h)	L5
42.0	5/5	5, 5, 2, 5, 2	44, 47, 24, 47, 22	3.14–3.17	0.072–0.072
44.2	5/5	5, 3, 5, 5, 2	47, 36, 44, 47, 26	3.17–3.17	0.072–0.072
46.4	5/5	5, 5, 4, 5, 2	44, 55, 52, 47, 25	3.17–3.17	0.072–0.072
48.7	5/5	5, 2, 5, 5, 2	47, 26, 44, 57, 28	3.17–3.17	0.072–0.072
50.9	5/5	5, 5, 2, 5, 2	44, 59, 30, 55, 28	3.17–3.17	0.072–0.072
53.1	5/5	5, 2, 5, 5, 3	58, 30, 44, 61, 44	3.17–3.17	0.072–0.072
55.3	5/5	5, 5, 2, 5, 2	54, 63, 32, 61, 32	3.17–3.17	0.072–0.072
57.6	5/5	5, 3, 5, 5, 2	63, 46, 56, 65, 34	3.17–3.17	0.072–0.072
59.8	5/5	5, 5, 2, 5, 1	60, 67, 36, 67, 23	3.04–3.20	0.072–0.087
62.0	5/5	4, 2, 4, 4, 1	57, 36, 53, 57, 25	3.17–3.22	0.072–0.072

```
**4. Risk to probe**
- The 0.0 prices on V_ant, V_hap, V_loop, and V_ferr hide the cost of the cuts. The antenna lost 533.303 from
    its claim, and the record doesn't show it still performs.
- The record also never states the achieved dB at 1 m. Only p_siren (0.814), pmarg (1.5), and the 3.0 floor
    are given. Verify the 86 dB requirement directly.
```

Opus 5.5 (290 words, 54/54 grounded):

```
# Decision Record: Volume and Area Allocation

**1. Concessions versus original claims**

These domains gave ground:
- Structure: V_struct 1800 -> 1400.
- Antenna: V_ant 1500 -> 1029.74.
- Acoustics: V_acou 1460 -> 1209.036, split 0.78 -> 0.3, r_mesh 40 -> 20.
- Haptics: V_hap 350 -> 294.615.
- NFC: V_loop 320 -> 225.66, V_ferr 180 -> 155.659.
- Board: A_board 1250 -> 1197.0.

These variables gained:
- Battery: V_batt 3786 -> 3858.882, A_batt 800 -> 807.149.
- Other: t_wall 1.2 -> 1.263, port 9 -> 9.454, h_out 1.0 -> 3.5, A_spr 320 -> 320.761.

**2. Couplings that drove the concessions**

- **Battery volume** was the costliest coupling to settle: bat:V_batt = -2333.04. The siren floor follows at
    ac:P_siren = -1232.65.
- **Acoustic volume** was the next costliest: ac:V_acou = -927.74, then str:t_wall = -918.25 and board:A_board
     = -869.35.
- **Port** was contested from both sides with opposite signs: str:port = +288.73 against ac:port = -283.43.
- **Antenna and NFC** conceded almost without resistance: ant:V_ant = 0.11, nfc:V_ferr = 0.0, nfc:V_loop =
    0.0.
```

```
**3. What the design gives up**

- The siren sits exactly at its floor: 3.0 achieved against a floor of 3.0, with no margin.
- The other objectives settled at runtime 60.93, voice 75.52 and l5 0.096.
- Six constraints are binding or near-binding: C1_volume, batt_t_max, O6_pressure, O5_skin_temp, haptic_force
    and board_routability. That leaves essentially no slack for growth.

**4. Risk to probe**

The 86 dB @ 1 m requirement is not reported among the outputs. Only proxies appear: p_siren 0.87, pmarg 1.5,
    q_ss 1.691 and f_res 3490.695.

Reviewers should demand direct verification of the 86 dB requirement. The siren is pinned at its floor, and ac
    :P_siren carries the second-largest price, so any acoustic shortfall has no buffer.
```

Haiku 5.5 (247 words, 62/63 grounded):

```
**Decision trace**

**1. Concessions vs. original claims.** Largest cuts: V_struct 1800->1400 (-22%), V_ant 1500->1112.6 (-26%),
    V_acou 1460->1268.9 (-13%), V_batt 3786->3656.2, V_loop 320->228.7 (-29%), V_hap 350->294.6 (-16%),
    A_board 1250->1197, r_mesh 40->20 (-50%), split 0.78->0.3 (-62%). Relaxed upward: h_out 1.0->3.5, port
    9.0->10.749, t_wall 1.2->1.304, A_spr 320->321.249. A_batt held (800->799.733).

**2. Driving couplings.** By price magnitude: bat:A_batt (1.75e22), str:port (-5.01e21), board:A_board (-3.12
    e21), hap:V_hap (1.65e21), th:A_spr (-7.10e20), nfc:V_loop (-1.90e20). These map to the board, haptic,
    thermal and loop concessions. The voice/acoustic trade is driven by ac:voice (-9.09e19) and ac:V_acou
    (1.86e19). The V_struct cut is large, yet str:V_struct (1.29e19) is far smaller than the top prices, so
    it is not the dominant driver; this needs explaining.

**3. What the design gives up.** L5 reaches only 0.11 despite weight 0.3. Voice reaches 76.378 with the
    acoustic budget reduced. Split and r_mesh are cut by 62% and 50%. Siren meets its floor of 3.0 exactly,
    with no margin.

**4. Risk to probe.** Dual prices span about 1e17 to 1e22, and the largest sits on bat:A_batt, where the
    allocation barely moved (799.733 vs 800). Either the duals are poorly scaled or that constraint is
    degenerate, so price rankings should not be trusted until they are normalized. Also, the data gives no dB
     value for the siren. The 86 dB @ 1 m and >=3 h requirements rest on the model outputs (p_siren 0.814,
    f_res 3490.695 Hz), and the siren sits exactly on its floor. Verify SPL and runtime against measurement
    or higher-fidelity modelling before sign-off.
```

S8 Data Sources

- Platform envelope, mass, water and shock ratings, siren level, and battery capacity: the manufacturer's published specification page. The source is withheld to keep the platform anonymous and is available to the editor on request.
- Driver: Same Sky CMS-18132-18SP and CMS-18138A-SP microspeaker datasheets, `https://www.sameskydevices.com/product/resource/cms-18132-18sp.pdf` and `https://www.sameskydevices.com/product/resource/cms-18138a-sp.pdf`.
- Acoustic vents: W. L. Gore & Associates acoustic vent design guidance (GAW346 class) for the membrane architecture studied during model development; the main case uses an open cavity with a mesh of at least 20 rayl.
- Speech Intelligibility Index: ANSI/ASA S3.5-1997 (R2017) [1].
- Touch-temperature limit: IEC 62368-1:2018, Clause 9.2.6, Table 38 [2], and [3].
- Antenna bound: Chu [4].

S9 Software and Reproducibility

Python 3.13, NumPy 2.5.3, SciPy 1.18.1, pymoo 0.6.2, MOSQITO 1.2.1, Anthropic Python SDK 2.3.0. Models: `claude-sonnet-5-5`, `claude-haiku-5-5`, `claude-opus-5-5`, provider default sampling (the SDK exposes no temperature parameter for these models). One design evaluation takes about 0.5 ms on a single CPU core.

The accompanying code archive contains the benchmark schema (`space_claim_schema_v0.7.json`), evaluators, the five methods, analysis scripts, and all run outputs. The main entry points are:

```
python3 arm2v7_run.py 0,1,2,3,4          # A2, five seeds
python3 arm1_sweep.py                      # A1, 50 programs
python3 arm1_pm.py 0; python3 arm1_pm.py 1  # A1-PM, 50 programs with PM escalation
python3 arm3_sweep.py                      # A3, 35 weightings (+ restoration)
python3 arm4v2_run.py; python3 arm4v2_run_rep.py r2; python3 arm4v2_run_rep.py r3   # A4
python3 arm4pm_run.py balanced,battery_first,...                                    # A4-PM
python3 arm5_final.py r1                   # A5 probe + grounded formulation (one-sided engine)
python3 arm5_final2.py r1                  # A5 split-floor engine on saved formulations
python3 arm5_xmodel.py <model> r1          # complete A5 pipeline for one model
python3 arm5_records_v2.py <model> <run.json> <out.json>   # decision records, 4000-token limit
python3 arm_xmodel_compare.py; python3 make_figs.py         # tables and figures
```

The API key is read from a local file or environment variable and is not included.

S10 Protocol Changes During the Study

The following changes were made before the reported runs unless stated otherwise. They are listed so that the history of the protocol is visible.

1. **Volume pool.** An early schema omitted the NFC loop from the volume total, understating over-commitment; corrected to 10.3%.
2. **Board/battery stack.** An area-based constraint made the shipped configuration infeasible and was replaced by the Z-stack constraint.
3. **Acoustic model.** An uncalibrated damping index was replaced by the physical circuit (Section S3); a membrane architecture was studied and set aside because its stiffness made the siren requirement infeasible; the voice proxy was replaced by the SII.
4. **A1 fairness.** Acoustics was first forced to fill all remaining space, which raises siren power; it now chooses its own volume.
5. **A4 pilot.** A pilot exposed missing what-if probes on shared inputs, silent clipping, and an ambiguity about whether the perimeter keepout lay inside the interior volume. All were fixed before the first sweep. The first sweep (8 directions) exposed ledger-only acceptance and stale acceptance, fixed by P1 and P2 before the reported 30 negotiations.
6. **A5 history.** A zero-shot formulation (A5a), a variant in which the LLM also set coordination parameters (A5b), and a variant with those parameters locked (A5c) preceded the grounded pipeline. A5b collapsed convergence (0 of 45 strictly feasible), which led to removing solver parameters from the catalog. Placing the siren floor at the arbiter stalled coordination at about 0.2% residual; placing it only in acoustics (A5d) made it soft; the split floor (A5e) is the reported engine.
7. **Formulation reuse.** The Sonnet split-floor runs re-solved the three formulations written in the A5d runs rather than requesting new ones; the Haiku and Opus pipelines formulated afresh.
8. **PM variants (added to represent industry practice).** A1-PM and A4-PM were added so that the domain-level processes are also tested with the program-manager arbitration used in practice. During development of A1-PM, levers E3 and E4 were added after E1–E2 alone plateaued below 3 h in 30 of 50 programs; the reported runs use all four. The first A4-PM attempt stopped when API credit ran out and a second at a checkpointing bug; neither produced results, and the reported ten negotiations are a complete fresh run.

9. **Decision records (after the reported runs).** The records were first generated with a 1200-token output limit, which the models' internal reasoning also consumed; most were cut short, and the Sonnet split-floor records had been generated without the program-direction text. All 90 records were regenerated with a 4000-token limit and the direction text, from the saved designs and solver data. All 90 completed. Only the record statistics changed.

References

[1] Acoustical Society of America, *Methods for Calculation of the Speech Intelligibility Index*, ANSI/ASA S3.5-1997 (R2017).

[2] International Electrotechnical Commission, *Audio/video, information and communication technology equipment—Part 1: Safety requirements*, IEC 62368-1:2018.

[3] S. Yoo *et al.*, "Responsive materials and mechanisms as thermal safety systems for skin-interfaced electronic devices," *Nat. Commun.*, vol. 14, 2023, doi: 10.1038/s41467-023-36690-y.

[4] L. J. Chu, "Physical limitations of omni-directional antennas," *J. Appl. Phys.*, vol. 19, no. 12, pp. 1163–1175, 1948.

[5] M. R. Stinson, "The propagation of plane sound waves in narrow and wide circular tubes, and generalization to uniform tubes of arbitrary cross-sectional shape," *J. Acoust. Soc. Am.*, vol. 89, no. 2, pp. 550–558, 1991.